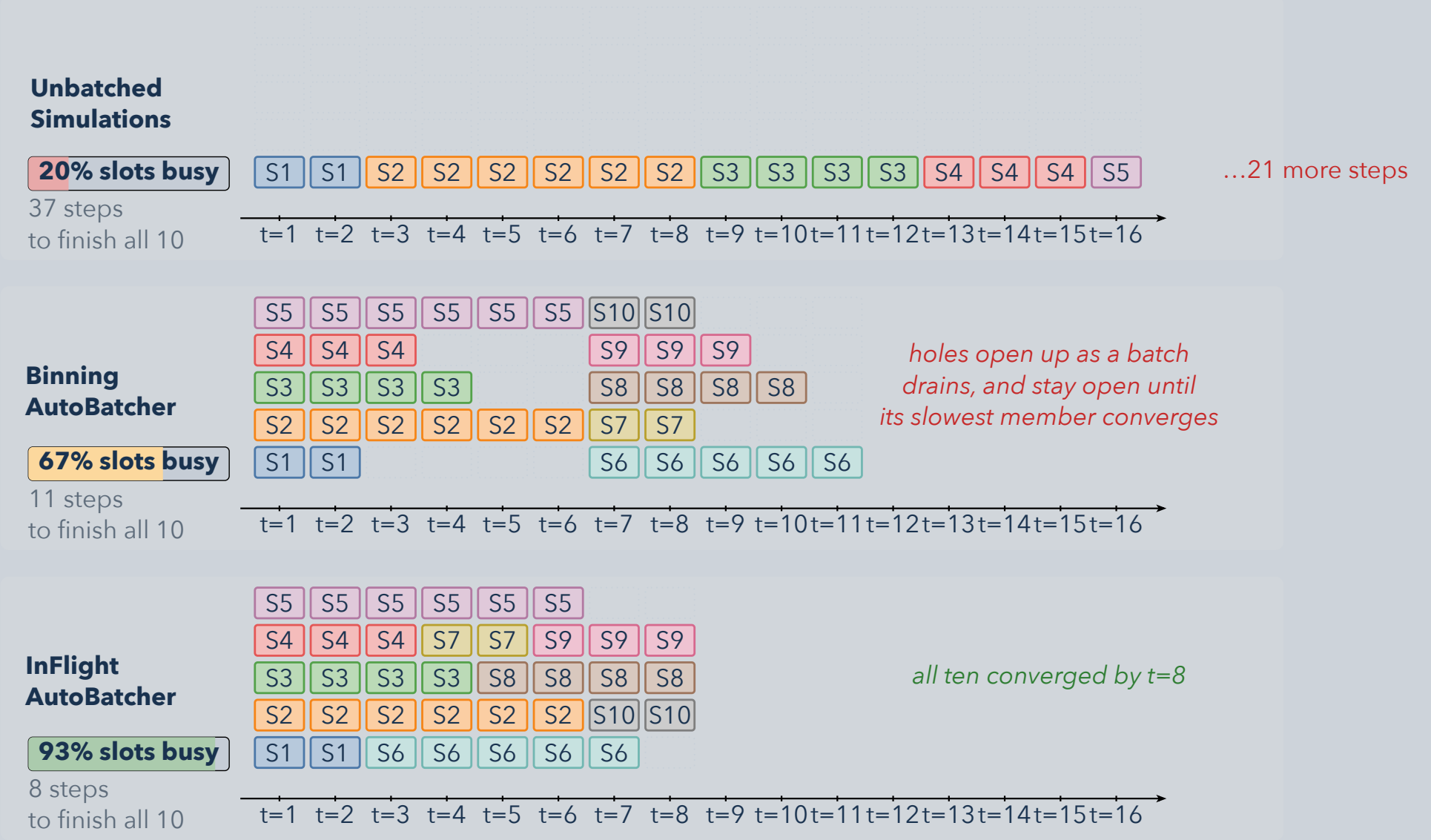


Small atomistic jobs can leave a GPU underused. Batching runs several together; replacing completed jobs keeps useful work in the batch.

1 Compare scheduling strategies

Ten structures needing 2, 6, 4, 3, 6, 5, 2, 4, 3, 2 ionic steps, on 5 GPU slots



Unbatched leaves four of five slots idle. **Binning** holds each slot until the batch’s slowest member converges. **In-flight** immediately reuses freed slots: a color change within a row marks a new structure taking over.

2 Follow a replacement

